

The family index of the odd signature operator with coefficients in a flat bundle

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Abstract

The Atiyah-Singer index theorem provides an analytic interpretation of certain topological invariants associated to a M -bundle $f : E \rightarrow B$, M a smooth closed oriented Riemannian manifold. In particular a family of such operators $D = \{D_b\}_{b \in B}$ produces a class $\text{index}(D) \in K^*(B)$ in the complex K-theory of B .

For example, the index of the (even) signature operator D^e coincides with the signature $\text{Sig}(M)$ of the intersection form defined on the cohomology of M ,
 $\text{index}(D^e) = \text{Sig}(M) = \langle \mathcal{L}(M), [M] \rangle \in \mathbb{Z} \cong K^0(pt.)$, where $\mathcal{L}(M) = \mathcal{L}(TM)$ is the total Hirzebruch L-class of M .

Any elliptic operator D on a manifold M may be “twisted” by a vector bundle $V \rightarrow M$, producing an operator D_V . When V is *flat* $\text{index}(D_V^e)$ again may be interpreted as the signature of an intersection form, this time defined on the *cohomology* $H^*(M; V)$ of M with coefficients in V ,

$$\text{index}(D_V^e) = \text{Sig}_V(M) = \langle \mathcal{L}(M) \text{ch}([V]), [M] \rangle \in \mathbb{Z} \cong K^0(pt.).$$

Sig_V are the *higher signatures*, conjectured by S. Novikov to be invariants of the oriented homotopy type of M .

Assume M is odd dimensional. There is also an odd signature operator D^o and the family index of D^o is still related to the signature of E . It was shown by J. Ebert that the family index of D^o is always zero, $\text{ind}(D^o) = 0 \in K^1(B)$, from which it follows that $\text{Sig}(E) = 0$. Ebert’s proof, essentially functional analytic in nature, is still applicable to D_V^o , so long as V is flat Hermitian. Let $V \rightarrow E$ be a flat hermitian vector bundle.

Theorem. The the family index of the odd signature operator twisted by V is zero,
 $\text{ind}(D_V^o) = 0 \in K^1(B)$.

Let $f_! : H^*(E) \rightarrow H^{*-\dim(M)}(B)$ be the umkehr homomorphism (integration over the fiber), $T_v E = \ker(df)$ the vertical tangent bundle of E . Applying the Chern character $\text{ch} : K \rightarrow H\mathbb{Q}$ yields the following vanishing results in rational cohomology.

Theorem. $f_!(\mathcal{L}(T_v E) \text{ch}([V])) = 0 \in H^*(B; \mathbb{Q})$.

Theorem. The higher signature $\text{Sig}_V(E) = 0$.

The special case when $V = 0$ was conjectured by Atiyah and proven by J. Ebert. This result generalizes an observation made by M. Gromov (when $M = S^1$). Note the flatness of V above is essential. This can be seen in the simplest possible example where $E = S^1 \times S^1 \rightarrow B = S^1$ is the projection.

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